

-
- 

DEVPOST

-  Search hackathons... 
- - [Log in](#)
 - [Sign up](#)
- 

[Join a hackathon](#) 



[Devpost](#)

[Participate in our public hackathons](#)

[Hackathons](#) → [Projects](#) →



[Devpost for Teams](#)

[Access your company's private hackathons](#)

[Login](#) →

[Host a hackathon](#) 



[Devpost](#)

[Grow your developer ecosystem and promote your platform](#)

[Host a public hackathon](#) →



[Devpost for Teams](#)

[Drive innovation, collaboration, and retention within your organization](#)

[Host an internal hackathon](#) →

By use case

[All hackathons](#) → [Customer hackathons](#) → [Employee hackathons](#) → [Public hackathons](#) →

[Resources](#) 

[Blog](#)

[Insights into hackathon planning and participation](#)

[Customer stories](#)

[Inspiration from peers and other industry leaders](#)

[Planning guides](#)

[Best practices for planning online and in-person hackathons](#)

[Webinars & events](#)

[Upcoming events and on-demand recordings](#)

[Help desk](#)

[Common questions and support documentation](#)





- [Join a hackathon](#) ▾ ▴

- [Back](#)



- [Devpost](#)

- [Participate in our public hackathons](#)

- [Hackathons](#) → [Projects](#) →



- [Devpost for Teams](#)

- [Access your company's private hackathons](#)

- [Login](#) →

- [Host a hackathon](#) ▾ ▴

- [Back](#)



- [Devpost](#)

- [Grow your developer ecosystem and promote your platform](#)

- [Host a public hackathon](#) →



- [Devpost for Teams](#)

- [Drive innovation, collaboration, and retention within your organization](#)

- [Host an internal hackathon](#) →

- By use case

- [AI hackathons](#) → [Customer hackathons](#) → [Employee hackathons](#) → [Public hackathons](#) →

- [Resources](#) ▾ ▴

- [Back](#)

- [Blog](#)

- [Insights into hackathon planning and participation](#)

- [Customer stories](#)

- [Inspiration from peers and other industry leaders](#)

- [Planning guides](#)

- [Best practices for planning online and in-person hackathons](#)

- [Webinars & events](#)

- [Upcoming events and on-demand recordings](#)

- [Help desk](#)

- [Common questions and support documentation](#)



-

- [Log in](#)



- [Sign up](#)

Neuthera - Drug discovery toolkit

NeuThera is an AI-driven drug discovery toolkit integrating multiple SOTA generative models for de novo molecular design. Combines graph-based retrieval with advanced compound fingerprint embeddings.

[Like 434](#)
[Comment 9](#)

- [Story](#)
- [Updates](#)



- >
- 1
 - 2
 - 3

NeuThera - AI-Driven Drug Discovery Toolkit

Introduction

Traditional drug discovery methods are hindered by high costs, long timelines upwards of 20 years, and a high attrition rate in clinical trials. With the rise of AI and computational biology -- Generative models and in-silico validation techniques have shown promise in accelerating early-stage drug design. However, current methods often operate independently of each other, lacking an integrated pipeline for end-to-end drug candidate generation.

NeuThera was proposed as a modular drug discovery starter toolkit, integrating multiple state-of-the-art generative molecular design models such as [TamGen](#), [MolVec](#), and [DeepPurpose](#) with other advanced tooling like graph-based retrieval, molecular fingerprint embeddings, ADMET profiling, etc.

What it does

NeuThera is to create a fully modular agentic AI framework capable of generating, validating, and optimizing novel drug candidates in an end-to-end pipeline -- and maintaining a graph database of all FDA approved and generated compounds for further repurposing or discovery.

NeuThera supports both **end-to-end drug discovery automation** and **interactive, user-guided drug design workflows**.

Generate de novo molecules: using multiple SOTA generative models, including Transformer-based molecular generator models. Ability to generate millions of compounds with complete virtual ADMET profiling for validation. All generated compounds are viable in the real world.

Medical knowledge-base: Combines 15+ datasets from BioSNAP, DrugBank, ChEMBL, and PDB to create a starting point for drug discovery and repurposing.

Molecular Vector Embedding Space: Vector embeddings from chemBERTa facilitate lead optimization, scaffold hopping, and similarity search.

In Silico Testing: With generated compounds and target proteins, uses DeepPurpose to predict binding affinity without requiring real world intervention.

How we built it

Neuthera is architected as a multi-model system, integrating specialized tools and frameworks to create a starter toolkit for end-to-end drug discovery. The modular architecture and the technology used allows researchers to expand the tooling, as all necessary data and functions are already provided.

Core Modules

- **TamGen:** A transformer-based generative model for de novo molecular design, trained on 20 million+ SMILES of known bioactive compounds. All generated compounds are validated and docked on target protein using DockBox before final result.
- **DeepPurpose:** A multi-model framework used to check the binding affinity between a drug and a target protein. As all generated compounds from TamGen are already validated with docking coordinates, DeepPurpose fits in perfectly with this pipeline.
- **chemBERTa:** Transformer-based model trained on 120+ million SMILES to create meaningful vector embeddings. Used to create vector space of all FDA Approved drugs + generated compounds for advanced analytics.

Knowledge Graph & Retrieval Modules

- **ArangoDB:** A multimodal database storing structured relationships between genes, proteins, diseases, drugs, and molecular structures.
- **Graph-based querying** for retrieving relevant molecular information, linking known drug-target interactions to novel compounds.

Molecular Similarity & Lead Optimization

- **Morgan/Circular Fingerprints:** Used for molecular embedding, leveraging Tanimoto similarity and vector representations.
- **Vector Space Search:** for identifying structurally similar compounds within large-scale chemical libraries.
- **Graph-based lead optimization:** Using molecular graphs and chemical property analysis to refine generated compounds.

4. GraphRAG & Biomedical Reasoning

- **GraphRAG framework** Augments traditional RAG (Retrieval-Augmented Generation) methods by incorporating graph-based reasoning for biomedical data
- **Ontological mapping:** Integrates structured biomedical ontologies (e.g., DrugBank, ChEMBL) to enhance retrieval and inference.

Challenges we ran into

- **Data inconsistencies and scattered datasets:** As the biomedical data space is vast and varied from field to field (Most use different identifiers), it required us multiple days to put together the graph database using only open-source mappings. A lot of the datasets used required combining multiple different mappings to get the required fields.
- **Outdated libraries and models:** As most of the technology we're using is state of the art, they are not mature enough to work well with different technologies. Example: TamGen only has around 30 stars on github and required very specific library versions to work. We had to rewrite some of the code to work for our needs and combine with our tech stack. Sadly, we've had to discard nx-arangodb as it does not function well with TamGen which is a core unit of our project.
- **Scaling the solution:** As college students with almost no resources, we've had to keep our AI model (GPT-4o) usage to a bare-minimum. As such, we could not fine tune it to work as we intended, though it works well ~75% of the time. We wish to work on this project for a long and with adequate funding, we can run a model with much better reasoning abilities and fine tune to be extremely useful.

Accomplishments that we're proud of

- **TamGen code rewrite to work without CUDA:** Our code rewrite for TamGen can be a valid contribution for their repository.
- **Biomedical knowledge retrieval:** Combining 15+ datasets enables us to do structured hypothesis-driven drug design.
- **Scalable molecular similarity search:** using vector embeddings for rapid lead identification and generated drug repurposing.

What we learned

- We got a glimpse of how cutting edge research for generative AI works.
- How vast and wonderful the open source community for biomedical fields are.
- Working with multiple tooling and creating coherence between them.
- Integrating SOTA technologies like ArangoDB and GraphRAG to create meaningful toolings

What's Next for NeuThera?

Enhancing generative models with a feedback loop:

- Integrating diffusion-based generative modeling to improve the novelty, drug-likeness, and synthetic accessibility of generated compounds.
- Implementing a reinforcement-driven feedback loop for iterative optimization
 1. **TamGen** generates an initial batch of 10000 de novo compounds.
 2. **DeepPurpose** predicts binding affinity for each compound with the target protein.
 3. Compound encoders select the highest-scoring compounds as reference for next batch.
 4. Loop continues until convergence, maximizing binding affinity while optimizing ADMET properties.

Automated lead optimization and drug repurposing:

- Integrating Graph Neural Networks (GNNs) on molecular embeddings to predict novel indications for existing compounds.
- Scaffold hopping via vector space traversal to explore chemically diverse, high-affinity derivatives.

Modular tooling for user-defined extensions:

- Allowing users to integrate custom generative models, scoring algorithms, functions, etc.
- Supporting custom scoring functions, including multi-objective optimization for ADMET, pharmacokinetics, and toxicity filtering.

Advanced UI/UX for visualization and interactive drug design:

- Developing an interactive web platform with:
 - AI-Chatbot system
 - Molecular visualization for ligand-protein interactions and generated compounds.
 - Graph-based relationship exploration to analyze drug-target-disease links.
 - Live compound generation and scoring dashboard, enabling real-time feedback on molecular design.



With

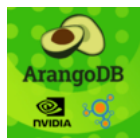
ngo
[mart](#)
python
snap
tgpt
mberta
[mbi](#)
ppurpose
gbank
ss
phrag
ggingface
gchain
[das](#)

- [python](#)
- [rdkit](#)
- [tamgen](#)
- [torch](#)

Try it out

- [GitHub Repo](#)

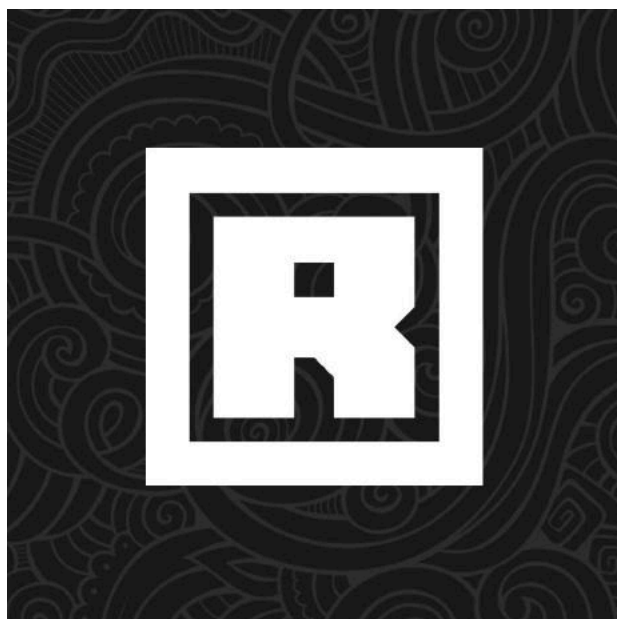
Submitted to



[Building the Next-Gen Agentic App with GraphRAG & NVIDIA cuGraph](#)

- Winner Third Place

Created by



[Jones David](#)



[praneeth yeddu](#)



[Like 434](#)

434 people like this:



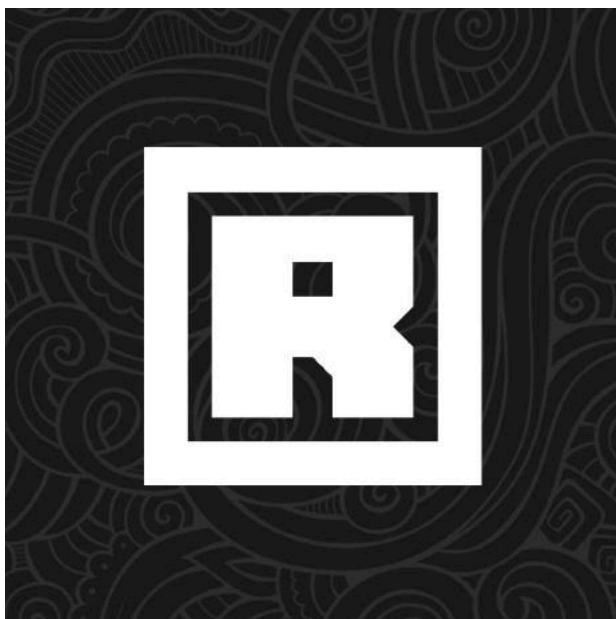


+ 424 more

Share this project:



Updates



Jones David started this project — about a year ago

Leave feedback in the comments!

[View previous comments](#)



Private user · 9 months ago

Greetings, Barsham, ⚠ Major flaws in the logic and design of the project

1. Ambiguity in the definition of the model's goal

The project goal is "predicting the site and location of polyadenylation", but it is not clear whether the model is supposed to:

- predict the exact site (regression or localization),
- or just classify the presence/absence of APA (binary classification).

This ambiguity in the type of model output affects the entire architectural design and evaluation.

2. Inconsistency in the selection of features and the model

The DNABERT-2-117M model is sequence-based, but many of the selected features (such as distance to the stop codon, location, gene_id) are non-sequence-based and structural. This conflict between the data type and the model type can cause a model to use or ignore structural information.

3. Data reduction without quality control

During the training phase, the data volume is halved "to save time," without any mention of the sample selection method. This can result in the removal of key samples, reduced motif diversity, and bias in the model. Without quality control, this reduction is dangerous.

4. Lack of clear separation between humans and mice

It is mentioned that PolyASite and PolyA_DB include human and mouse data, but it is not clear whether the model was trained on both species or only one. Combining different species without separation can be misleading and produce unreliable outputs.

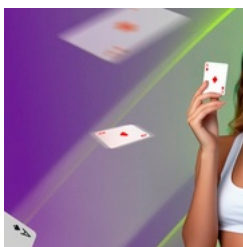
5. Ambiguity in evaluation and accuracy metrics

The evaluation section only mentions the use of transformers and scikit-learn, but no specific features (such as F1, ROC-AUC, or confusion matrix) for the performance of the model are mentioned. Without precise metrics, the model cannot be evaluated.

6. Over-reliance on visualization tools

Using BertViz to visualize the model's attention is useful, but if the model is not properly trained or the data is incomplete, this training can be misleading and provide visual beauty rather than scientific analysis.

In the meantime, for machine learning you need this



[Mateslots Casino](#) · 9 months ago

Good idea!



[Toqeer Ahmad](#) · 7 months ago

This is an incredibly impressive and ambitious project! NeuThera's modular approach to AI-driven drug discovery especially the integration of TamGen, chemBERTa, and DeepPurpose within a unified pipeline really highlights how generative AI can revolutionize biomedical research. The feedback-loop idea for iterative compound optimization is brilliant. Excited to see how this evolves into a fully deployable toolkit!



[EZHIL ARASAN RAMANAN](#) · 6 months ago

Extremely impressed. Can we connect at my personal mail id? drpharmaindia@gmail.com





[goldendoodle-breeders](#) · 3 months ago

This discussion is really insightful! I especially appreciate the detailed critique about model goals, feature selection, and evaluation metrics it's a reminder of how careful design and clear definitions are crucial in AI projects. It's also exciting to see ideas like NeuThera's AI-driven drug discovery pipeline showing the potential of generative models in biomedical research.

On a lighter note, when I need a small mental break from dense research topics, I like visiting The Doodle Tribe <https://share.google/CM3b9sbdENvWeZXA3>, a website about doodle dogs and companion puppies. It's a fun way to relax and recharge before diving back into complex projects!

[Log in](#) or [sign up for Devpost](#) to join the conversation.

Devpost

- [About](#)
- [Careers](#)
- [Contact](#)
- [Help](#)

Hackathons

- [Browse hackathons](#)
- [Explore projects](#)
- [Host a hackathon](#)
- [Hackathon guides](#)

Portfolio

- [Your projects](#)
- [Your hackathons](#)
- [Settings](#)

Connect

- [Twitter](#)
- [Discord](#)
- [Facebook](#)
- [LinkedIn](#)

© 2026 Devpost, Inc. All rights reserved.

- [Community guidelines](#)
- [Security](#)
- [CA notice](#)
- [Privacy policy](#)
- [Terms of service](#)